Assignment 1　　杜奥旋　　122090095

1. $\because X_i \sim \chi^2(r_i)$ $\therefore f(x_i)=\frac{1}{\Gamma(\frac{r_i}{2})\cdot 2^{\frac{r_i}{2}}} x_i^{\frac{r_i}{2}-1} e^{-\frac{x_i}{2}}$ $M(t)=(1-2t)^{-\frac{r_i}{2}}$

$Y=\sum_{i=1}^{n} X_i$ $M_Y(t)=\prod_{i=1}^{n} M_{x_i}(t)=(1-2t)^{\frac{-(r_1+\cdots+r_n)}{2}}$ $\therefore Y\sim\chi^2(\sum r_i)$

2. $P(X=x_i)=\frac{\lambda^{x_i}e^{-\lambda}}{x_i!}$, $\mu=1.8$, $\sigma^2=1.8$

flooded: $f(x)=\frac{1}{\theta}e^{-\frac{x}{\theta}}$, $\mu=3$, $\sigma^2=9$

(a) $P(X=x)=\frac{\lambda^x e^{-\lambda}}{x!}$ $\lambda=\mu=1.8\times 20=36$

$P(X\geq 19)=1-\sum_{x=1}^{18}\frac{36^x e^{-36}}{x!}=0.9993$ $\mu=36$, $\sigma^2=36$, $\sigma=6$

by CLT

$P(X\geq 19)$

$Z=\frac{X-\mu}{\sigma}=\frac{X-36}{6}\geq\frac{-17}{6}\approx -2.83$

$P(Z\geq -2.83)\approx 1-0.0023=0.9977$

(b) $P(Y<365)$ $\mu=120\times 3=360$ $\sigma^2=1080$ $Y\sim N(360,1080)$

$Z=\frac{Y-\mu}{\sigma}<\frac{365-360}{\sqrt{1080}}=0.1521$

$P(Z<0.152)\approx 0.5596$

3. (a) $\hat{\lambda}_{mom}=\frac{1}{\bar{X}}$

(b) $L(\lambda)=\prod_{i=1}^{n}\lambda e^{-\lambda x_i}$ $f_X(x)=\begin{cases}\lambda e^{-\lambda x}, & \text{if } x>0\\ 0, & \text{otherwise}\end{cases}$

$l(\lambda)=n\ln\lambda-\lambda\sum_{i=1}^{n}x_i=n\ln\lambda-n\lambda\bar{x_i}$

$l'(\lambda)=\frac{n}{\lambda}-n\bar{x_i}=0$ $\hat{\lambda}=\frac{1}{\bar{x_i}}$ $\therefore \hat{\lambda}_{mle}=\frac{1}{\bar{x_i}}$

(c) $f(x|\lambda)=\lambda e^{-\lambda x}$ $\mu=\frac{1}{\lambda}=\bar{x}=\frac{4.8+3.25+1.21+0.64+2.51+4.39}{6}=2.8$

$\therefore \lambda_{mom}=\frac{1}{2.8}\approx 0.3571$

$\lambda_{mle}=\frac{1}{\bar{x_i}}=\frac{1}{2.8}\approx 0.3571$

(d) $E(\lambda_{mle})=E(\frac{n}{\sum x_i})=\frac{n}{E(\sum x_i)}=\frac{n}{n\cdot\frac{1}{\lambda}}=\lambda$

$\therefore \lambda_{mle}$ is an unbiased estimator of λ.

4. $\int_a^b \frac{x}{b-a}\,dx = \frac{1}{b-a}\int_a^b x\,dx = \frac{b+a}{2}$

$f(x)=\frac{1}{b-a}$, $\mu=\frac{a+b}{2}$, $\sigma^2=\frac{(a-b)^2}{12}$

$$M(t)=\begin{cases}\frac{e^{tb}-e^{ta}}{t(b-a)}, & t\neq 0\\ 1, & t=0\end{cases}$$

$\frac{\hat{a}+\hat{b}}{2}=\bar{X}$ $\quad a=2\bar{x}-b$

$\frac{(\hat{a}-\hat{b})^2}{12}=\frac{1}{n}\Sigma x^2-\bar{x}^2=E(X^2)-[E(X)]^2$

$\hat{a}=2\bar{x}-\hat{b}$

$$\therefore \begin{cases}\hat{b}=\frac{3\Sigma x^2}{n\bar{x}}-2\bar{x}\\ \hat{a}=4\bar{x}-\frac{3\Sigma x^2}{n\bar{x}}\end{cases}$$

5. $$f_X(x)=\begin{cases}e^{-(x-\theta)}, & \text{if } x\geq 0\\ 0, & \text{otherwise}\end{cases}$$

$L(\theta)=e^{-\Sigma x+n\theta}$

$l(\theta)=-\Sigma x+n\theta \qquad l'(\theta)=n>0$

$\therefore L(\theta)$ is an increasing function of θ for $\theta\in X_i$

Suppose that $X_{(1)}=\min\{x_1,x_2,\dots,x_n\}$ $\Rightarrow$ MLE is $\hat{\theta}=X_{(1)}$

6. $X\sim U(0,\theta)$

$f(x)=\frac{1}{\theta}$, $0\leq x\leq\theta$ $\quad L(\theta)=\frac{1}{\theta^n}$ $\quad \max\left(\frac{1}{\theta^n}\right)\Leftrightarrow\min(\theta^n)$, subject to $\theta\geq\max(x)$, $\theta\in\min(x)$

$\therefore \hat{\theta}=\max(X_1,X_2,\dots,X_n)$

cdf: $$F(x)=\begin{cases}0, & x<0\\ \frac{x}{\theta}, & 0\leq x<\theta\\ 1, & x\geq\theta\end{cases}$$ $\quad E(X)=\frac{\theta}{2}$

suppose $\hat{X}=\max\{X_1,X_2,\dots,X_n\}$ $\quad P(\hat{X}\leq x)=P[X_1\leq x, X_2\leq x,\dots,X_n\leq x)=P(X_1\leq x)\cdot P(X_2\leq x)\cdots P(X_n\leq x)$

$\therefore$ for every X_i, $P(X_i\leq x)=\frac{x}{\theta}$

$$\therefore P(\hat{X}\leq x)=\begin{cases}0, & x<0\\ \left(\frac{x}{\theta}\right)^n, & 0\leq x\leq\theta\\ 1, & x>\theta\end{cases}$$

$E(\hat{X})=\int_0^\theta x\cdot\left(\frac{1}{\theta^n}\right)\cdot n x^{n-1}\,dx=\frac{n}{n+1}\theta$

$\therefore \hat{X}$ is biased

$\hat{\theta}_{\text{unbiased}}=\frac{n+1}{n}\hat{X}$

$E(\hat{\theta}_{\text{unbiased}})=\frac{n+1}{n}E(\hat{X})=\theta$ $\quad \therefore \frac{n+1}{n}\hat{X}$ is an unbiased estimator of θ.

7 (a) $P(X,K) = P(X|K)P(K)$

$\therefore P(K=0) = \pi_0$, $P(K=1) = \pi_1 = 1-\pi_0$

$P(X|K=0) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(X-\mu_0)^2}{2\sigma_0^2}\right)$, $P(X|K=1) = \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(X-\mu_1)^2}{2\sigma_1^2}\right)$

$\therefore P(X,K) = \begin{cases} \pi_0 \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(X-\mu_0)^2}{2\sigma_0^2}\right), & K=0 \\ \pi_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(X-\mu_1)^2}{2\sigma_1^2}\right), & K=1 \end{cases}$

(b) $L(\pi_0, \mu_0, \sigma_0^2, \mu_1, \sigma_1^2 | X, K) = \prod_{i=1}^{n_0} \left(\pi_0 \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(x_i-\mu_0)^2}{2\sigma_0^2}\right)\right) \prod_{i=1}^{n_1} \left(\pi_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(x_i-\mu_1)^2}{2\sigma_1^2}\right)\right)$

$l = \sum_{i=1}^{n_0} \left[\log(\pi_0) - \frac{1}{2}\log(2\pi\sigma_0^2) - \frac{(x_i-\mu_0)^2}{2\sigma_0^2}\right] + \sum_{i=1}^{n_1} \left[\log(1-\pi_0) - \frac{1}{2}\log(2\pi\sigma_1^2) - \frac{(x_i-\mu_1)^2}{2\sigma_1^2}\right]$

$\frac{dl}{d\pi_0} = \frac{n_0}{\pi_0} - \frac{n_1}{1-\pi_0} = 0 \quad \therefore \pi_0 = \frac{n_0}{n_1}(1-\pi_0)$

$\frac{n}{\pi_0} = \frac{n_1}{1-\pi_0} \quad \hat{\pi}_0 = \frac{n_0}{n_0+n_1}$

$\frac{dl}{d\mu_0} = \sum_{i=1}^{n_0} \frac{x_i-\mu_0}{\sigma_0^2} = 0 \quad \therefore \hat{\mu}_0 = \frac{1}{n_0}\sum_{i=1}^{n_0} x_i$

$\frac{dl}{d\sigma_0^2} = -\frac{n_0}{2\sigma_0^2} + \sum_{i=1}^{n_0} \frac{(x_i-\mu_0)^2}{2\sigma_0^4} = 0 \quad \therefore \hat{\sigma}_0^2 = \frac{1}{n_0}\sum_{i=1}^{n_0} (x_i-\hat{\mu}_0)^2$

$\frac{dl}{d\mu_1} = \sum_{i=1}^{n_1} \frac{x_i-\mu_1}{\sigma_1^2} = 0 \quad \therefore \hat{\mu}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} x_i$

$\frac{dl}{d\sigma_1^2} = -\frac{n_1}{2\sigma_1^2} + \sum_{i=1}^{n_1} \frac{(x_i-\mu_1)^2}{2\sigma_1^4} = 0 \quad \therefore \hat{\sigma}_1^2 = \frac{1}{n_1}\sum_{i=1}^{n_1} (x_i-\hat{\mu}_1)^2$

C)

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import numpy as np
import pandas as pd

# Load the dataset from GMM.csv
data = pd.read_csv("GMM.csv")

# Separate data into groups based on K
group_0 = data[data['K'] == 0]
group_1 = data[data['K'] == 1]

# Calculate sample means and variances for each group
mu_0 = group_0['X'].mean()
sigma_0 = group_0['X'].var()
mu_1 = group_1['X'].mean()
sigma_1 = group_1['X'].var()

# Calculate sample proportions for pi_0 and pi_1
n_0 = len(group_0)
n_1 = len(group_1)
n_total = len(data)
pi_0 = n_0 / n_total
pi_1 = n_1 / n_total

# These are the MLEs for the parameters
print("MLE for pi_0:", pi_0)
print("MLE for mu_0:", mu_0)
print("MLE for sigma^2_0:", sigma_0)
print("MLE for mu_1:", mu_1)
print("MLE for sigma^2_1:", sigma_1)
```

result:

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MLE for pi_0: 0.78006
MLE for mu_0: 19.977871852721485
MLE for sigma^2_0: 100.59798751142192
MLE for mu_1: 39.98854741964292
MLE for sigma^2_1: 25.273565682084474
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